4. Environmental Engineering :

4.1 Water supply—Predicting demand for water, impurities, of water and their significance, physical, chemical and bacteriological analysis, waterborne diseases, standards for potable water.

4.2 Intake of water—Water treatment—principles of coagulation, flocculation and sedimentation; slow-; rapid-, pressure-, filters; chlorination, softening, removal of taste, odour and salinity.

4.3 Sewerage systems—Domestic and industrial wastes, storm sewage–separate and combined systems, flow through sewers, design of sewers.

4.4 Sewage characterization—BOD, COD, solids, dissolved oxygen, nitrogen and TOC. Standards of disposal in normal watercourse and on land.

4.5 Sewage treatment—Working principles, units, chambers, sedimentation tanks, trickling filters, oxidation ponds, activated sludge process, septic tank, disposal of sludge, recycling of wastewater.

4.6 Solid waste—Collection and disposal in rural and urban contexts, management of long-term ill effects.

5. Environmental pollution—Sustainable development. Radioactive wastes and disposal. Environmental impact assessment for thermal power plants, mines, river valley projects. Air pollution. Pollution control acts.

COMMERCE & ACCOUNTANCY

PAPER–I

Accounting & Finance

Accounting, Taxation & Auditing :

1. Financial Accounting—Accounting as a Financial Information System; Impact of Behavioural Sciences. Accounting Standards *e.g.*, Accounting for Depreciation, Inventories, Research and Development Costs, Long-term Construction Contracts, Revenue Recognition, Fixed Assets, Contingencies, Foreign Exchange Transactions, Investments and Government Grants, Cash Flow Statement, Earnings Per Share.

Accounting for Share Capital Transactions including Bonus Shares, Right Shares, Employees Stock Option and Buy- Back of Securities.

Preparation and Presentation of Company Final Accounts.

Amalgamation, Absorption and Reconstruction of Companies.

2. Cost Accounting—Nature and Functions of Cost Accounting. Installation of Cost Accounting System. Cost Concepts related to Income Measurement, Profit Planning, Cost Control and Decision Making.

Methods of Costing: Job Costing, Process Costing, Activity Based Costing.

Volume–cost–Profit Relationship as a tool of Profit Planning.

Incremental Analysis/Differential Costing as a Tool of Pricing Decisions, Product Decisions, Make or Buy Decisions, Shut-Down Decisions etc.

Techniques of Cost Control and Cost Reduction: Budgeting as a Tool of Planning and Control. Standard Costing and Variance Analysis.

Responsibility Accounting and Divisional Performance Measurement.

3. Taxation—Income Tax : Definitions; Basis of Charge; Incomes which do not form Part of Total Income. Simple problems of Computation of Income (of Individuals only) under Various Heads, *i.e.*, Salaries, Income from House Property, Profits and Gains from Business or Profession, Capital Gains, Income from other sources, Income of other Persons included in Assessee's Total Income .

Set-Off and Carry Forward of Loss. Deductions from Gross Total Income. Salient Features/Provisions Related to VAT and Services Tax.

4. Auditing—Company Audit : Audit related to Divisible Profits, Dividends, Special investigations, Tax audit.

Audit of Banking, Insurance, Non-Profit Organizations and Charitable Societies/Trusts/Organizations.

Financial Management, Financial Institutions and Markets :

1. Financial Management—Finance Function : Nature, Scope and Objectives of Financial Management: Risk and Return Relationship.

Tools of Financial Analysis : Ratio Analysis, Funds-Flow and Cash-Flow Statement.

Capital Budgeting Decisions : Process, Procedures and Appraisal Methods. Risk and Uncertainty Analysis and Methods.

Cost of capital: Concept, Computation of Specific Costs and Weighted Average Cost of Capital. CAPM as a Tool of Determining Cost of Equity Capital.

Financing Decisions : Theories of Capital Structure - Net Income (NI) Approach, Net Operating Income (NOI) Approach, MM Approach and Traditional Approach. Designing of Capital structure: Types of Leverages (Operating, Financial and Combined), EBIT-EPS Analysis, and other Factors.

Dividend Decisions and Valuation of Firm: Walter's Model, MM Thesis, Gordan's Model Lintner's Model. Factors Affecting Dividend Policy.

Working Capital Management : Planning of Working Capital. Determinants of Working Capital. Components of Working Capital-Cash, Inventory and Receivables.

Corporate Restructuring with focus on Mergers and Acquisitions (Financial aspects only)